

Meta-Learning-Driven Joint Power Allocation and Phase-Shift Design for RIS-Assisted NOMA-ISAC Systems

First Author, *Graduate Student Member, IEEE*, Second Author, *Member, IEEE*, and
Third Author, *Senior Member, IEEE*

Abstract—This letter investigates a reconfigurable intelligent surface (RIS)-assisted multiple-input single-output (MISO) integrated sensing and communication (ISAC) system, where a base station (BS) simultaneously serves a near and a far user via non-orthogonal multiple access (NOMA) while illuminating a sensing target. We maximize a weighted communication-and-sensing throughput by jointly optimizing the NOMA power-allocation coefficients and the RIS phase shifts, subject to an aggregate communication QoS floor, individual near/far-user QoS floors, and a sensing QoS constraint. To tackle the resulting non-convex problem, a model-agnostic meta-learning (MAML)-based deep-learning algorithm is proposed. A lightweight network predicts the low-dimensional power split, while an inner optimizer adapts the high-dimensional RIS phase vector for each channel realization. A first-order meta-update then trains the power map using the phase-adapted channels, avoiding the unstable higher-order derivatives observed in this high-SINR setting. Simulation results show stable convergence, a strong RIS-aperture gain, and consistent improvements against OMA, no-RIS, and partially optimized baselines.

Index Terms—Reconfigurable intelligent surface (RIS), integrated sensing and communication (ISAC), non-orthogonal multiple access (NOMA), deep learning, meta-learning (MAML), MISO.

I. INTRODUCTION

INTEGRATED sensing and communication (ISAC) has emerged as a key enabler for next-generation wireless networks, jointly designing sensing and communication on a shared spectral and spatial resource to improve spectrum, energy, and hardware efficiency [1], [2]. When combined with NOMA, which superposes users in the power domain and separates them by successive interference cancellation (SIC), ISAC can further raise spectral efficiency. Severe blockage and path loss, however, degrade both functions, motivating the use of an RIS to reshape the propagation environment through low-cost passive elements [3].

RIS-assisted ISAC resource management has attracted growing interest. In [4], hybrid beamforming and RIS phase shifts were jointly optimized to shape the ISAC beampattern. In [5], a penalty-based block-coordinate method minimized the Cramer-Rao bound for RIS-aided mmWave ISAC. A rate-splitting RIS-ISAC design was studied in [6]. For RIS-NOMA networks, [7] developed a DL/MAML framework that jointly optimizes BS

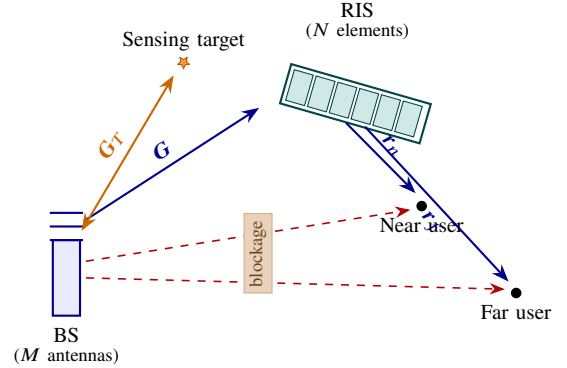


Fig. 1. System model: a MISO BS serves a near and a far user via NOMA with an N -element RIS while sensing a point target.

power allocation and RIS phase shifts. These results motivate a learning-based factorization of the coupled phase-and-power design: the low-dimensional power allocation is predicted by a neural network, whereas the high-dimensional RIS phase is refined per channel realization.

The main contributions are summarized as follows. First, we formulate a weighted communication-plus-sensing throughput maximization problem containing both a cluster-level communication QoS floor and individual near/far-user QoS floors, so aggregate spectral efficiency cannot hide user starvation. Second, we develop a first-order MAML-based bilevel algorithm in which the power split is predicted from the current phase-dependent channel features and the RIS phase is adapted by a short per-channel inner optimization. Third, we discuss the algorithmic convergence limitations and the offline/online computational cost, and provide a reproducible simulation table containing the actual link distances, fading/path-loss settings, RIS size, power, and learning parameters.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. System Model

As illustrated in Fig. 1, a BS equipped with M antennas serves a single-antenna near user (n) and far user (f) and illuminates a point sensing target (T). The RIS contains N passive reflecting elements. Following the standard RIS model in [7], [8], the coefficient of element i is

$$v_i = \beta_i e^{j\phi_i}, \quad 0 \leq \beta_i \leq 1, \quad \phi_i \in [0, 2\pi), \quad (1)$$

and

$$\Phi = \text{diag}(\beta_1 e^{j\phi_1}, \dots, \beta_N e^{j\phi_N}). \quad (2)$$

For an ideal passive RIS we set $\beta_i = 1$ for all i , as commonly adopted to maximize reflected power [7], [8]; hence only the phases are optimized.

1) *Channel Model*: Let $\mathbf{G} \in \mathbb{C}^{N \times M}$ denote the BS-RIS channel and $\mathbf{r}_k \in \mathbb{C}^N$ the RIS-user channel. The cascaded channel of user $k \in \{n, f\}$ is

$$\mathbf{h}_k = \mathbf{G}^H \Phi \mathbf{r}_k \in \mathbb{C}^M. \quad (3)$$

All links use distance-dependent large-scale attenuation and small-scale fading. In the RIS-aided case the direct BS-user links are blocked and (3) dominates; the no-RIS baseline retains only the weak direct link. The round-trip sensing response is represented by $\mathbf{G}_T \in \mathbb{C}^{M \times M}$ and can be written for a point target as

$$\mathbf{G}_T = \sqrt{\beta_T} (\mathbf{G}^H \Phi \mathbf{h}_{RT}) (\mathbf{h}_{TR}^H \Phi \mathbf{G}), \quad (4)$$

where β_T is the round-trip target/path gain (40 dB in the simulator) and $\mathbf{h}_{RT}$ and $\mathbf{h}_{TR}$ denote the RIS-target and target-RIS links. This rank-one sensing form and the use of transmit/receive beamformers are consistent with RIS-assisted ISAC models such as [4], [5].

2) *NOMA Communication and Sensing Model*: The BS transmits

$$\mathbf{x} = (\sqrt{a_n P} s_n + \sqrt{a_f P} s_f) \mathbf{w}_c + \sqrt{a_T P} \mathbf{w}_T s_T, \quad (5)$$

where P is the total BS power and $\mathbf{a} = [a_n, a_f, a_T]$ contains the near-user, far-user, and sensing power fractions. The coefficients satisfy $a_n + a_f + a_T = 1$ and $a_k \geq 0$. The beamforming vectors $\mathbf{w}_c$ and $\mathbf{w}_T$ are unit-norm communication and sensing beamformers, respectively. Beamforming notation of this form is standard in RIS-ISAC resource allocation [4], [8].

For power-domain NOMA, both users share the common beam $\mathbf{w}_c$ and are separated by SIC [7], [12]. In the implemented fair-NOMA configuration, the communication beam is a normalized weighted combination

$$\mathbf{w}_c = \frac{0.7 \mathbf{h}_f + 0.3 \mathbf{h}_n}{\|0.7 \mathbf{h}_f + 0.3 \mathbf{h}_n\|}, \quad (6)$$

which deliberately gives the weaker far user the larger steering weight while retaining near-user gain. The symbols $\mathbf{w}_c$ and $\mathbf{w}_T$ therefore denote unit-norm active transmit beamforming vectors at the BS, whereas β_i is the passive amplitude coefficient of RIS element i ; these quantities play different roles and should not be conflated. Beamforming/RIS notation of this form is standard in RIS-NOMA/ISAC models [7], [4].

For sensing, define $\mathbf{H}_u = [\mathbf{h}_n, \mathbf{h}_f]$. The implemented sensing beam is the dominant eigenvector of the leakage-penalized Hermitian matrix

$$\mathbf{A}_T = \mathbf{G}_T^H \mathbf{G}_T - \lambda_{\text{null}} \mathbf{H}_u \mathbf{H}_u^H, \quad \lambda_{\text{null}} = 0.5, \quad (7)$$

followed by unit-norm normalization. This trades target illumination against communication leakage, consistent with leakage-aware ISAC beam design [4], [5].

With $g_{kj} = |\mathbf{h}_k^H \mathbf{w}_j|^2$, the SINRs are

$$\gamma_f = \frac{a_f g_{ff} P}{a_n g_{ff} P + a_T g_{fT} P + \sigma^2}, \quad (8)$$

$$\gamma_n = \frac{a_n g_{nn} P}{a_T g_{nT} P + \sigma^2}, \quad (9)$$

$$\gamma_s = \frac{a_T |\mathbf{u}_T^H \mathbf{G}_T \mathbf{w}_T|^2 P}{\sigma^2}, \quad (10)$$

where σ^2 is the noise power and $\mathbf{u}_T$ is the sensing receive combiner. The achievable rates are $R_k = \log_2(1 + \gamma_k)$, $k \in \{n, f, s\}$.

The weighted system rate is

$$R_{\text{DL}} = \omega_c (R_n + R_f) + \omega_s R_s, \quad \omega_c + \omega_s = 1. \quad (11)$$

The communication rates are summed in the *objective* because the near and far users occupy the same NOMA resource block, so $R_n + R_f$ is the cluster spectral efficiency. The weights ω_c and ω_s are design priorities that trade communication throughput against sensing performance; in our simulations they are fixed to 0.7 and 0.3, respectively. Importantly, using a sum in the objective does not replace individual QoS protection.

B. Problem Formulation

We jointly optimize the power split $\mathbf{a}$ and RIS phases $\phi = [\phi_1, \dots, \phi_N]$. The communication sum rate is retained as a cluster-level QoS measure, while individual user constraints are added in the same spirit as the per-device QoS constraint in (9b) of [7]. The resulting problem is

$$(P1) \quad \max_{\mathbf{a}, \phi} \quad R_{\text{DL}} \quad (12a)$$

$$\text{s.t.} \quad R_n + R_f \geq R_{\text{th},c}, \quad (12b)$$

$$R_k \geq R_{\text{th},k}, \quad k \in \{n, f\}, \quad (12c)$$

$$R_s \geq R_{\text{th},s}, \quad (12d)$$

$$a_n + a_f + a_T = 1, \quad a_n, a_f, a_T \geq 0, \quad (12e)$$

$$|e^{j\phi_i}| = 1, \quad i = 1, \dots, N. \quad (12f)$$

The aggregate constraint (12b) is useful because the two NOMA users share one time-frequency resource block, so $R_n + R_f$ directly measures the communication service delivered by the cluster. It also matches the cluster-level QoS term used by the implemented training loss. On its own, however, a sum threshold is insufficient: a very large near-user rate could compensate for an unacceptable far-user rate. Constraint (12c) therefore adds the individual protection requested in the formulation, exactly analogous to writing a QoS condition for each device as in (9b) of [7]. Thus, the sum threshold controls aggregate communication quality while the individual thresholds prevent user starvation. The sensing requirement is enforced separately by (12d). The ideal-RIS assumption $\beta_i = 1$ is fixed rather than optimized, and the phase parameterization $e^{j\phi_i}$ enforces unit modulus automatically.

Problem (P1) remains non-convex because the power variables and RIS phases are coupled through the cascaded channels. We therefore enforce the QoS conditions through penalty terms during learning, while the softmax output enforces the power simplex exactly.

III. PROPOSED MAML-BASED SOLUTION

The implementation follows a bilevel power/phase decomposition. The three-dimensional power vector is predicted by a compact neural network, while the N RIS phases are adapted for each channel realization. This follows the basic motivation of [7]: learn the lower-dimensional resource map and refine the channel-specific RIS configuration with gradient-based adaptation.

A. Feature Map and Power-Allocation Network

For a given phase vector, the cascaded channels and the six effective gains $\{g_{nn}, g_{ff}, g_{sT}, g_{fn}, g_{fT}, g_{nT}\}$ are computed. They are converted to dB and concatenated with the communication/sensing QoS levels and the SNR budget, giving a nine-dimensional input feature vector $\mathbf{z}$. The power network contains three hidden layers with 256 neurons per layer, LayerNorm and LeakyReLU activations, followed by a three-neuron softmax head. Hence

$$\mathbf{a} = \text{softmax}(f_{\theta}(\mathbf{z})), \quad (13)$$

which enforces $a_n + a_f + a_T = 1$ and $a_k \geq 0$ by construction.

B. Joint Power-Allocation and Phase Adaptation

For a channel sample and the current phase vector $\phi^{(j)}$, the network first predicts

$$\mathbf{a}^{(j)} = \text{softmax}\left(f_{\theta}(\mathbf{z}(\phi^{(j)}))\right). \quad (14)$$

The implemented constrained loss is

$$\begin{aligned} \ell^{(j)} = & -R_{\text{DL}} + \lambda_c (R_{\text{th},c} - R_n - R_f)_+ + \lambda_s (R_{\text{th},s} - R_s)_+ \\ & + \lambda_n (R_{\text{th},n} - R_n)_+ + \lambda_f (R_{\text{th},f} - R_f)_+, \end{aligned} \quad (15)$$

where $(x)_+ = \max(x, 0)$. Equation (15) makes explicit why the aggregate and individual communication thresholds coexist: λ_c protects cluster throughput, whereas λ_n and λ_f penalize starvation of either user.

The RIS phase is represented by $\theta_i = e^{j\phi_i} = \cos \phi_i + j \sin \phi_i$, so unit modulus is satisfied for any real ϕ_i . For each channel, the phase vector is randomly initialized and adapted for K steps using Adam on (15),

$$\phi^{(j+1)} = \text{Adam}_{\phi}\left(\phi^{(j)}, \nabla_{\phi^{(j)}} \ell^{(j)}; \eta_{\phi}\right), \quad (16)$$

with inner learning rate η_{ϕ} . Since the effective gains change after every phase update, the feature vector and power split are recomputed at each inner step.

C. First-Order Meta-Update

After the K phase-adaptation steps, the final rates and loss are recomputed at $\phi^{(K)}$. The implementation uses a first-order MAML approximation: the adapted phase is treated as fixed during the outer network update rather than backpropagating second-order derivatives through the entire Adam trajectory. The outer update is therefore

$$\theta \leftarrow \theta - \alpha \nabla_{\theta} \frac{1}{B} \sum_{b=1}^B \ell_b(\theta, \phi_b^{(K)}), \quad (17)$$

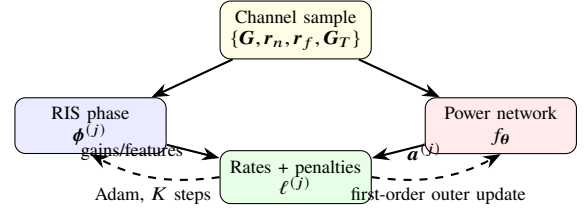


Fig. 2. First-order MAML-based joint power/phase design. The inner loop adapts the RIS phase and recomputes the network power split; the outer loop updates the power network using the phase-adapted channel while ignoring higher-order derivatives through the inner optimizer.

Algorithm 1: First-order MAML-based power and RIS-phase optimization

Input: channel data; inner steps K ; inner Adam rate η_{ϕ} ; outer rate α ; penalties $\lambda_c, \lambda_s, \lambda_n, \lambda_f$.

Output: trained power-map parameters θ^* .

Initialize θ

while not converged do

 Sample a mini-batch of B channels

foreach channel b **do**

 Initialize $\phi_b^{(0)} \sim \mathcal{U}([0, 2\pi)^N)$

for $j = 0$ **to** $K - 1$ **do**

 Compute phase-dependent channels/gains and $\mathbf{z}_b^{(j)}$

 Predict $\mathbf{a}_b^{(j)}$ using (14)

 Evaluate R_n, R_f, R_s and (15)

 Update $\phi_b^{(j+1)}$ with inner Adam

end

 Recompute $\mathbf{a}_b^{(K)}$ and final loss at the adapted phase

end

 Update θ with Adam using the first-order gradient in (17)

end

Select θ^* by validation score

where B is the mini-batch size. This retains the key meta-learning idea—the power map is trained on channels after per-instance RIS adaptation—while avoiding the memory and numerical burden of the full higher-order graph. In the implemented fair, high-SINR configuration, the full second-order path was found to produce unstable/non-finite meta-gradients, which motivated the first-order variant.

D. Training and Prediction Procedure

Algorithm 1 summarizes the implementation. At inference, the same K -step phase adaptation is performed for an unseen channel and the final softmax output provides the power allocation. Thus prediction is not a pure one-shot neural-network call; it is a short learned-plus-optimization procedure that trades a small fixed online cost for substantially better phase adaptation.

E. Convergence Analysis

A global convergence guarantee is not claimed. The objective is non-convex, the unit-modulus RIS parameterization couples

all rates through the cascaded channel, and the hinge penalties in (15) are non-smooth exactly at the QoS boundaries. Moreover, as discussed for the related RIS-NOMA MAML method in [7], standard smooth MAML convergence theorems do not directly cover an architecture in which the inner and outer loops optimize different variables. Our first-order approximation further discards the higher-order dependence of the adapted phase on θ . Consequently, convergence is assessed empirically: the inner Adam loop is run for a fixed number of steps and the outer training curve is checked for a stable loss/rate plateau across the reported hyperparameter settings.

F. Computational Complexity

Let P_θ denote the power-network parameter count. With nine inputs, three 256-unit hidden layers, LayerNorm, and a three-output head, the implemented network contains approximately 1.36×10^5 trainable parameters. Computing the user cascades costs $O(MN)$, while forming the sensing matrix and its $M \times M$ leakage-aware eigenvector adds approximately $O(MN + M^3)$. Therefore, one inner prediction/adaptation step costs

$$O(P_\theta + MN + M^3), \quad (18)$$

and online prediction for one channel realization costs

$$O(K(P_\theta + MN + M^3)), \quad (19)$$

plus one final forward evaluation of the same order. For a batch of B samples, an offline training iteration scales as $O(BK(P_\theta + MN + M^3))$. Because the implemented outer update is first-order, it does not store a full second-order graph through all K inner steps, reducing memory and derivative cost relative to full MAML. This is the relevant complexity for *prediction*: one compact network plus a fixed, short RIS adaptation, rather than solving a fresh long alternating-optimization problem from scratch for every channel.

IV. NUMERICAL RESULTS AND ANALYSIS

Table I reports the parameters used by the fair NOMA-RIS simulator and the final learning/evaluation scripts. The channel generator uses 10,000 realizations with a 90/10 train/validation split. The node geometry is specified in the simulator by link distances rather than unique Cartesian coordinates; reporting these distances therefore reproduces the propagation model without inventing coordinates that are not present in the code.

The aggregate NOMA threshold in Table I comes from the fair NOMA dataset, where it was chosen to keep the cluster QoS active; the individual near/far floors are the defaults used by the training scripts. The OMA fair dataset uses its own calibrated aggregate/sensing thresholds ($R_{th,c} = 12$ and $R_{th,s} = 2$ bit/s/Hz). The service-rate outage thresholds stored by the MATLAB generators are diagnostic quantities and are not the same as the optimization penalties in (15).

The implementation records per-iteration loss and weighted rate for convergence monitoring, while model selection uses a held-out validation score. The present analysis therefore relies on the fixed-step inner optimizer, validation-based checkpointing, and the convergence limitations discussed in Section III-E rather than asserting a closed-form global guarantee.

TABLE I
SYSTEM, CHANNEL, AND LEARNING PARAMETERS.

Parameter	Value
BS antennas M	4
RIS elements N	8 nominal; {8, 16, 32, 64} in RIS-size sweep
Power-sweep RIS size	$N = 16$
Monte-Carlo realizations	10,000 (90% train, 10% validation)
BS-RIS distance d_{BR}	8 m
RIS-near distance d_{RD_n}	3 m
RIS-far distance d_{RD_f}	25 m
RIS-target / target-RIS	$d_{RT} = d_{TR} = 6$ m
Direct BS-near / BS-far refs.	$d_{BD_n} = 10$ m, $d_{BD_f} = 25$ m
Reference path gain PL_0	10^{-3} (−30 dB)
Path-loss exponents	2.2, 2.8, 2.8, 2.3, 2.3, 3.8, 3.8
Nakagami- m parameters	$m_{BR} = 2$, $m_{RD_n} = m_{RD_f} = 1$; $m_{RT} = m_{TR} = 3$, $m_{BD_n} = m_{BD_f} = 1$
Target gain β_T	40 dB
RIS amplitude β_i / efficiency	1 / 1 (ideal passive RIS)
Bandwidth / temperature / NF	1 MHz / 290 K / 6 dB
Generator power P_{tot}	40 dBm; trainer permits power override
RIS-size figure operating power	15 dBm
Power sweep	5, 10, 15, 20, 25 dBm
Initial NOMA split (a_n, a_f, a_T)	(0.20, 0.55, 0.25)
Communication beam weights	far/near = 0.7/0.3
Sensing leakage penalty λ_{null}	0.5
Objective weights (ω_c, ω_s)	(0.7, 0.3)
NOMA cluster/sensing QoS	$R_{th,c} = 21.7$, $R_{th,s} = 2.9$ bit/s/Hz
Individual training floors	$R_{th,n} = R_{th,f} = 1$ bit/s/Hz
Penalty weights ($\lambda_c, \lambda_s, \lambda_n, \lambda_f$)	(2, 50, 15, 20)
Power network	9–256–256–256–3, softmax output
NOMA-RIS result checkpoints	batch 64, lr 10^{-4} , 40 epochs, $K = 10$
OMA-RIS result checkpoints	batch 64, lr 10^{-4} , 40 epochs, $K = 5$
No-RIS result checkpoints	batch 128, lr 5×10^{-4} , 60 epochs
Inner phase optimizer	Adam, learning rate 0.1

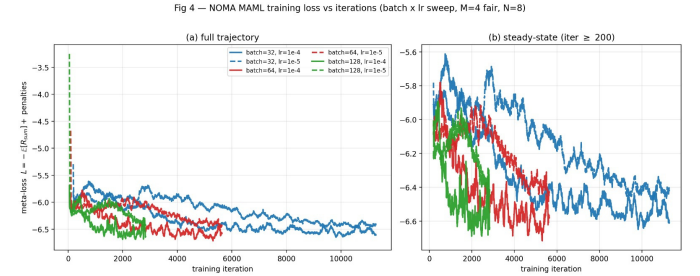


Fig. 3. MAML meta-loss versus training iteration over a batch/learning-rate sweep ($M=4$, $N=8$). (a) Full trajectory; (b) steady state (iter ≥ 200).

Fig. 3 shows the MAML meta-loss (15) over training iterations for several batch/learning-rate settings. The loss decreases smoothly and the weighted sum-rate climbs steadily, with no epoch-one collapse, indicating stable second-order meta-optimization; the steady-state panel confirms continued, generalizing improvement.

Fig. 4 plots the empirical CDF of the weighted sum-rate R_{DL} over the test set for the ablation. Joint optimization of phase and power (fullOptimized) stochastically dominates the partial designs, raising the median from ~ 5.0 to ~ 6.7 bps/Hz; phase optimization accounts for most of the rate gain, while the learned power split adds a further, QoS-critical improvement. This is reflected in Fig. 5: the joint QoS-violation rate falls monotonically from 29.4% (no optimization) to 0.1% (proposed), with both the communication (q_c) and sensing (q_s) constraints satisfied almost surely.

Fig. 6 reports R_{sum} versus the RIS size $N \in \{8, 16, 32, 64\}$ at $P = 15$ dBm. The rate grows monotonically with N for both

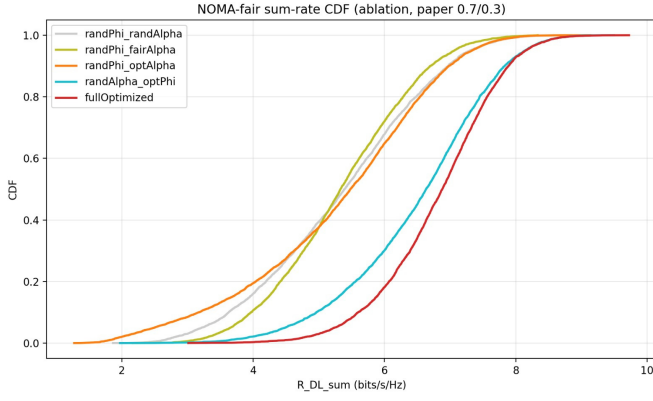


Fig. 4. CDF of the weighted sum-rate R_{DL} over the test set for the five ablation configurations.

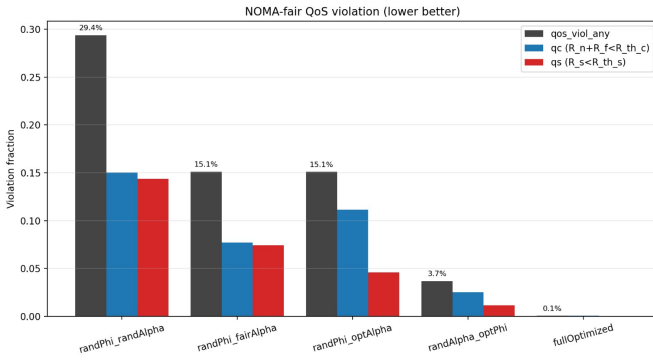


Fig. 5. QoS-violation fraction (lower is better). The joint violation rate drops from 29.4% to 0.1% along the ablation.

access schemes—from 6.9 to 11.4 bps/Hz for NOMA and from 4.1 to 7.3 bps/Hz for OMA—and NOMA retains an $\sim 65\%$ advantage throughout showing that the passive aperture and the access scheme contribute complementary gains.

Finally, Fig. 7 sweeps the transmit power. Panel (a) establishes the ordering $\text{NOMA-RIS} > \text{NOMA} > \text{OMA-RIS} > \text{OMA}$ across the whole range: NOMA’s full-time superposition outperforms OMA’s orthogonal slots, and the RIS adds a consistent margin on top of each access scheme (e.g. a $\sim 27\%$ RIS gain for NOMA at $P = 15$ dBm). Panel (b) shows the corresponding QoS-violation rate: NOMA-RIS reaches near-zero violation already at low power, whereas the no-RIS and OMA schemes need substantially more power to become feasible, quantifying the role of the RIS in jointly meeting the sensing and communication QoS.

Finally, two practical points follow. On *runtime*, the online stage is a single forward pass of f_θ plus $K=5$ light inner steps—orders of magnitude below per-instance AO—so the design fits the short coherence times that motivate learning-based ISAC. On *constraint granularity*, replacing the sum-rate floor (12b) with per-user penalties ($\lambda_n, \lambda_f > 0$ in (15)) lets the operator dial the near/far fairness trade-off, shifting the learned split toward the weaker user at a modest cost in aggregate rate and with no change to the pipeline.

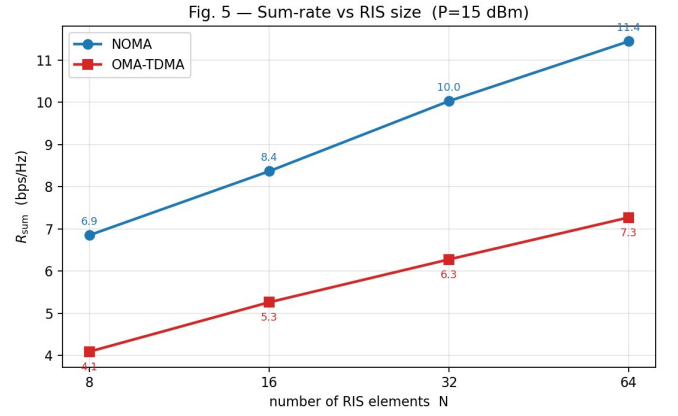


Fig. 6. Weighted sum-rate versus RIS size N at $P = 15$ dBm. NOMA dominates OMA at every aperture; both scale with N .

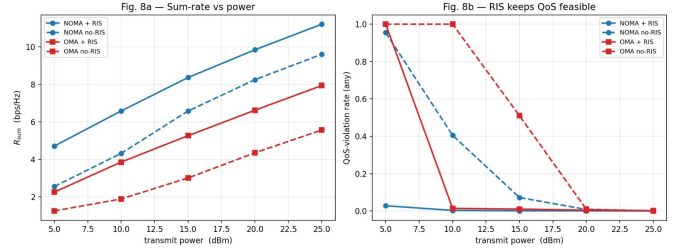


Fig. 7. (a) Weighted sum-rate and (b) joint QoS-violation rate versus transmit power for the four schemes.

V. CONCLUSION

A MISO RIS-assisted NOMA-ISAC downlink was investigated with joint power and phase design. The revised formulation combines an aggregate NOMA-cluster QoS constraint with individual near/far-user floors and a separate sensing floor, which preserves cluster efficiency while preventing a strong user from masking a weak one. The implemented solver uses a compact softmax power network and per-channel RIS-phase adaptation, followed by a first-order meta-update. The paper now reports the actual channel distances, fading/path-loss parameters, power/RIS settings, convergence limitations, and online prediction complexity. The original result figures and their surrounding discussion are retained and the figure files are provided in EPS form for submission.

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